

NAIVE

BAYES

$$P(\text{Yes}) = 9/14$$

$$P(\text{No}) = 5/14$$

$$P(\text{outlook} = \text{sunny} \mid \text{play} = \text{yes}) = \frac{2}{9}$$

$$P(\text{temperature} = \text{cold} \mid \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{humidity} = \text{high} \mid \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{wind} = \text{true} \mid \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{play} = \text{yes}) = \frac{9}{14}$$

$$P(\text{outlook} = \text{sunny} \mid \text{play} = \text{no}) = \frac{3}{5}$$

$$P(\text{temperature} = \text{cold} \mid \text{play} = \text{no}) = \frac{1}{5}$$

$$P(\text{humidity} = \text{high} \mid \text{play} = \text{no}) = \frac{4}{5}$$

$$P(\text{wind} = \text{true} \mid \text{play} = \text{no}) = \frac{3}{5}$$

$$P(\text{play} = \text{no}) = \frac{5}{14}$$

$$P(X | \text{play} = \text{yes}) \cdot P(\text{play} = \text{yes})$$

$$= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14}$$

$$= 0.0053$$

$$(P(X | \text{play} = \text{no}) \cdot P(\text{play} = \text{no}))$$

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14}$$

$$= 0.0206$$

$$P(X) = 0.0053 + 0.0206$$

$$= 0.0259$$

$$P(\text{play} = \text{Yes} | X) = \frac{P(X | \text{play} = \text{yes}) \cdot P(\text{play} = \text{yes})}{P(X)}$$

$$= 0.204$$

$$P(\text{play} = \text{No} | X) = \frac{P(X | \text{play} = \text{no}) \cdot P(\text{play} = \text{no})}{P(X)}$$

$$= 0.806$$

$\therefore$ Prediction that golf will be played is "No".